Firmware Upgrade & TCG Configuration Procedure (Target FW: 54T)

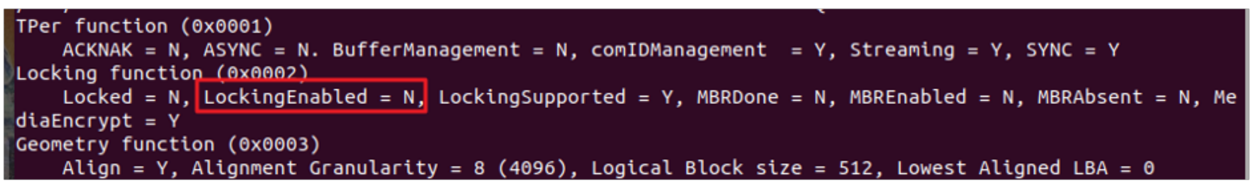
Step-by-Step Procedure

Step 1: Perform TCG Revert (Pre-Upgrade Preparation)
Reset the drive to factory default to eliminate potential upgrade risk caused by existing TCG configurations.

- Tool: Sedutil or customer's proprietary utility
- Action: Perform PSID Revert or SID Revert

Step 2: check the TCG revert
Confirm if TCG in manufactured-inactive state

- Use Sedutil: ./sedutil --query /dev/nvmex
- Then: The log shall show "LockingEnabled = N"



Step 3: Upgrade Firmware to Version 54T
Use the NVMe CLI tool to apply the firmware update.
Command:

- Run: nvme fw-download /dev/nvmex -f file.bin
- Then: nvme fw-commit /dev/nvmex -s 0 -a 3

Step 4: Verify Firmware Version
Confirm that the firmware has been successfully updated to 54T.
Command:

- Run: nvme list /dev/nvmex
- Check the reported firmware version.

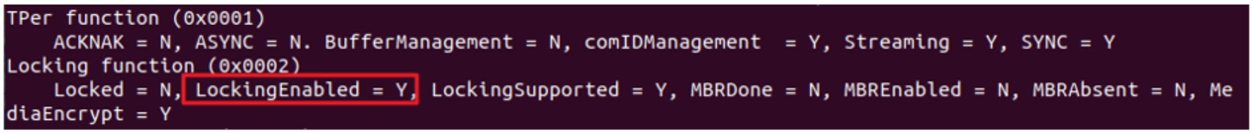
Step 5: Activate TCG Configuration
Initialize TCG settings post-upgrade.

- Use Sedutil: ./sedutil --initialSetup SID_new_password /dev/nvmex
- Alternatively, use the customer's tool to perform TCG activation.

Step 6: check the TCG activated status

Confirm if TCG in manufactured-active state

- Use Sedutil: ./sedutil --query /dev/nvmex
- Then: The log shall show “LockingEnabled = Y”



```
TPer function (0x0001)
    ACKNAK = N, ASYNC = N. BufferManagement = N, comIDManagement  = Y, Streaming = Y, SYNC = Y
Locking function (0x0002)
    Locked = N, LockingEnabled = Y, LockingSupported = Y, MBRDone = N, MBREnabled = N, MBRAbsent = N, Me
diaEncrypt = Y
```

Step 7: Perform Final TCG Revert (Post-Activation Cleanup)

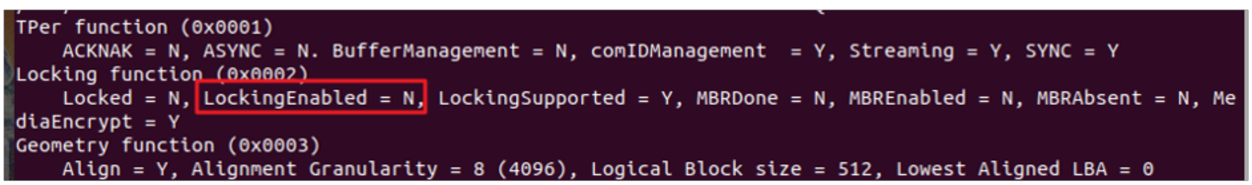
Ensure default TCG table settings from firmware 54T are properly applied.

- Tool: Sedutil or customer’s tool
- Action: Perform PSID Revert or SID Revert

Step 8: check the TCG revert status

Confirm if TCG in manufactured-inactive state

- Use Sedutil: ./sedutil --query /dev/nvmex
- Then: The log shall show “LockingEnabled = N”



```
TPer function (0x0001)
    ACKNAK = N, ASYNC = N. BufferManagement = N, comIDManagement  = Y, Streaming = Y, SYNC = Y
Locking function (0x0002)
    Locked = N, LockingEnabled = N, LockingSupported = Y, MBRDone = N, MBREnabled = N, MBRAbsent = N, Me
diaEncrypt = Y
Geometry function (0x0003)
    Align = Y, Alignment Granularity = 8 (4096), Logical Block size = 512, Lowest Aligned LBA = 0
```

Step 9: Begin Drive Usage & Validation

Proceed with normal drive operations and conduct functional testing as needed.

Procedure: Entering FIPS Mode

1. Ownership and Credential Setup

Take ownership of the Opal drive by setting the AdminSP SID credential to a value other than the default MSID.

- Action: Set a private value PIN of at least 8 bytes in length.

2. Locking SP Activation

The Drive Owner must execute the Activate method on the Locking SP.

- Action: Activate the Locking SP

3. Operator PIN Configuration

During installation, configure all applicable operator PINs (Drive Owner, Admins, and Users) to private values of at least 8 bytes.

- Action: Set a private value PIN of at least 8 bytes in length for all applicable operators.

4. Locking SP Admin Settings For TCG Opal Security Mode:

- Action: Set ReadLockEnabled and WriteLockEnabled to "**True**" on all data ranges.
- Action: These settings must not be modified after configuration.

Note: Two methods to verify the number of supported LockingRanges:

The first method: level 0 discovery

3.1.1.5 Opal SSC V2 Feature (Feature Code = 0x0203)

Table 7 Level 0 Discovery - Opal SSC V2 Feature Descriptor

Byte \ Bit	7	6	5	4	3	2	1	0
0	(MSB)	Feature Code (0x0203)						
1								(LSB)
2	Feature Descriptor Version Number				SSC Minor Version Number			
3	Length							
4	(MSB)	Base ComID						
5								(LSB)
6	(MSB)	Number of ComIDs						
7								(LSB)
8	Reserved for future common SSC parameters							Range Crossing Behavior
9	(MSB)	Number of Locking SP Admin Authorities Supported						
10								(LSB)
11	(MSB)	Number of Locking SP User Authorities Supported						
12								(LSB)
13	Initial C_PIN_SID PIN Indicator							
14	Behavior of C_PIN_SID PIN upon TPer Revert							
15-19	Reserved for future common SSC parameters							

4.2 Level 0 Discovery

An Opal-compliant SD that contains the Single User Mode feature set SHALL return the Single User Mode Feature Descriptor as described in 4.2.1, in addition to the Level 0 Discovery response requirements defined in [3].

4.2.1 Single User Mode Feature Descriptor (Feature Code = 0x0201)

This feature descriptor SHALL be returned when the Opal-compliant SD supports the Single User Mode feature set. The contents of the feature descriptor are defined in Table 7.

Table 7 Level 0 Discovery - Single User Mode Feature Descriptor

Bit / Byte	7	6	5	4	3	2	1	0
0	(MSB)	Feature Code						
1								(LSB)
2	Version				Reserved			
3	Length							
4-7	Number of Locking Objects Supported							
8	Reserved					Policy	All	Any
9-15	Reserved							

The second method : Get LockingInfo (Total number of supported ranges is **MaxRanges** + 1)

Table 45 Locking SP - LockingInfo Table Preconfiguration

UID	Name	Version	EncryptSupport	MaxRanges	MaxReEncryptions	KeysAvailableCfg	AlignmentRequired	LogicalBlockSize	AlignmentGranularity	LowestAlignedLBA
00 00 08 01 00 00 00 01			Media Encryption	8[1]						

5. Firmware Download Port Lock Configuration
    - Action: Set the value of Port LockOnReset for firmware download to "Power Cycle".
    - Action: This setting must not be modified after installation.

Firmware Upgrade Considerations for Drives in FIPS Approved Mode

If the drives remain in FIPS approved mode and the customer intends to upgrade firmware from version 54T to 550, the following settings must be reviewed and adjusted prior to initiating the upgrade:

Key Requirement:
Unlock Port LockOnReset

- In FIPS approved mode, the FW Download port has to remain locked which is the case when the LockOnReset value is set to **power-cycle** and the PortLocked value set to **True**.
- To proceed with the firmware upgrade, this The FW Download port needs to be **temporarily unlocked**.

Note: TCG OPAL FW Download Port Table

Col. 0 UID	1 Name	2 LockOnReset	3 PortLocked
00 01 00 02 00 01 00 02 (O)	FWDownload	[]	False

Verification Steps:

1. Check PortLocked Status:
   - Action: In FIPS approved mode, the PortLocked column value is "**True**".

2. **Before** Firmware Upgrade*:
   - Action: Set the **PortLocked** column value to "**False**" to allow firmware download and commit operations.

* Note: This is a critical step

3. **After** Firmware Upgrade:
   - Action: Set the **PortLocked** column value back to "**True**" to make sure the drive is locked against further update processes.

Firmware Upgrade Procedure: Version 550

Step-by-Step Procedure

Step 1: Upgrade Firmware to Version 550
Use the NVMe CLI tool to apply the firmware update.
Command:

- Run: nvme fw-download /dev/nvmex -f file.bin
- Then: nvme fw-commit /dev/nvmex -s 0 -a 3

Step 2: Verify Firmware Version
Confirm that the firmware has been successfully updated to 550.
Command:

- Run: nvme list /dev/nvmex
- Check the reported firmware version.